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Unit 4 - Dive into Flow

PDF Documents

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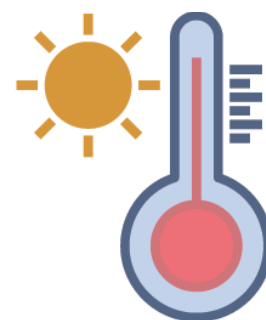
UNIT 4 DIVE INTO FLOW

Exploring Smart Devices Around Us

In today's world, smart devices have become a part of our daily lives, making things more efficient, convenient, and often safer. Beyond the smart wastebin, which opens automatically to help keep our hands clean, many other smart technologies are transforming how we interact with our environment. Here are a few common examples:

SMART THERMOSTATS

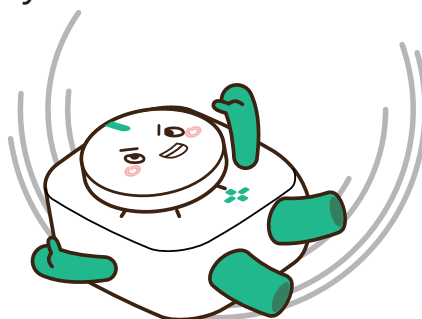
Smart thermostats learn your daily routine and adjust the temperature in your home accordingly. They help reduce energy consumption by ensuring your heating or cooling system works efficiently when needed. Many of them can also be controlled remotely via smartphones, so you can change the temperature before arriving home.



SMART LOCKS



Smart locks allow you to lock and unlock your doors remotely, using a smartphone or voice command. Some even allow access through fingerprint recognition or personalized codes. They provide enhanced security and convenience, especially for those who tend to lose their keys.



SMART SPEAKERS

Devices like Amazon Echo or Google Home are popular smart speakers that can play music, control other smart devices, answer questions, and set reminders—all through voice commands. They act as central hubs for managing various smart systems in your home.



SMART SECURITY CAMERAS



Smart security cameras allow you to monitor your home in real-time through your smartphone. They can alert you to movement, record footage, and even enable two-way communication, helping ensure your safety.

These smart devices improve convenience, reduce energy consumption, and promote safety in our homes and public spaces. As technology continues to evolve, we will likely see even more innovative solutions that seamlessly integrate into our daily routines, making life easier and more connected.





Multiple Choice Questions

1 What technology do smart bins use to know when they are full?

- ☐ A Magic
- ☐ B Sensors
- ☐ C Guessing
- ☐ D Lights

CORRECT ANSWER

☒ B

2 How do smart bins help city workers?

- ☐ A They make them work longer hours
- ☐ B They help plan better trash collection routes
- ☐ C They provide snacks
- ☐ D They create more trash

CORRECT ANSWER

☒ B

3 Which module can sense the distance?

- ☐ A ToF
- ☐ B Button
- ☐ C IMU
- ☐ D Speaker

CORRECT ANSWER

☒ A

●● Reflection Questions

- 1 What other types of devices or tasks in your home could benefit from being “smart” or automated? How would it improve your daily life?

- 2 In what ways do smart wastebins help reduce waste and improve cleanliness in cities?

- 3 How can sensors like the Time of Flight (ToF) sensor be used to automate tasks and make daily activities easier?